

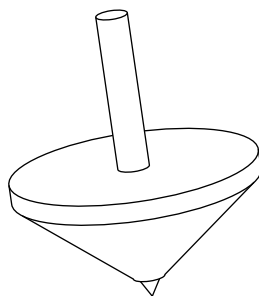
Do **not** write solutions on this page.

11. [Maximum mark: 14]

A mathematics class of 15 students plays a game which requires three equal size teams.

(a) Find the total number of ways that the three teams can be chosen. [3]

The game involves the spinning of a top.



The time, T , in minutes that the spinning top is in motion can be modelled by the probability density function f where

$$f(t) = \begin{cases} kte^{-3t}, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

and $k \in \mathbb{Z}^+$.

(b) Show that $\int_0^a f(t) dt = \frac{k}{9} [1 - (3a+1)e^{-3a}]$, where $a \in \mathbb{R}^+$. [4]

(c) (i) Use l'Hôpital's rule to find $\lim_{x \rightarrow \infty} (3x+1)e^{-3x}$.

(ii) Hence, by considering $\lim_{a \rightarrow \infty} \int_0^a f(t) dt$, find the value of k . [5]

(d) Find the median length of time that a spinning top is in motion. [2]

